
Session 5 — Advanced Probability & Statistics

Exercise sheet

This sheet is more demanding than the earlier ones. Each question tests a concept from the session, but backs it with a real calculation. Aim for short, precise answers that show both the number and its meaning.

Exercise 1 *Estimating a detection probability the Bayesian way.*

A detection probability θ has a prior $\text{Beta}(2, 2)$. You then observe 7 detections in 10 trials. (For a $\text{Beta}(a, b)$ prior and h successes in $h + t$ trials, the posterior is $\text{Beta}(a + h, b + t)$ with mean $\frac{a+h}{a+b+h+t}$.)

- (a) Give the posterior distribution and its mean.
 - (b) The prior mean was 0.5 and the data proportion was 0.7. In one sentence, how has the data moved your belief?
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Exercise 2 Continuing from the previous question, you observe 3 further detections in 4 new trials.

- (a) Update the posterior again, starting from $\text{Beta}(9, 5)$.
 - (b) Show that this equals what you would get by applying all $10 + 4$ trials at once to the original $\text{Beta}(2, 2)$ prior. What principle does this illustrate?
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Exercise 3 *A neuron provides evidence about whether a stimulus is a target.*

The prior odds that a stimulus is a target are $1 : 3$. Each firing of the neuron is 4 times more likely for targets than distractors (likelihood ratio 4), and successive responses are independent.

- (a) After one firing, compute the posterior odds and the posterior probability of a target.
 - (b) After a *second* firing, compute the new posterior probability.
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Exercise 4 Two binary variables — stimulus S (present/absent) and response R (yes/no) — have the joint distribution

$$\mathbb{P}(S=\text{pres}, R=\text{yes}) = 0.35, \quad \mathbb{P}(S=\text{pres}, R=\text{no}) = 0.05,$$

$$\mathbb{P}(S=\text{abs}, R=\text{yes}) = 0.10, \quad \mathbb{P}(S=\text{abs}, R=\text{no}) = 0.50.$$

- (a) Compute the marginals $\mathbb{P}(S=\text{pres})$ and $\mathbb{P}(R=\text{yes})$.
- (b) Compute the conditional $\mathbb{P}(S=\text{pres} \mid R=\text{yes})$.

(c) Are S and R independent? Justify with a calculation.

Exercise 5 *The covariance matrix is where statistics and linear algebra meet.*

A bivariate Gaussian has covariance matrix $\Sigma = \begin{pmatrix} 4 & 2 \\ 2 & 4 \end{pmatrix}$.

- (a) Find the eigenvalues of Σ .
 - (b) Give the direction of greatest variance and the fraction of total variance along it.
 - (c) Are the two variables independent? Explain.
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Exercise 6 Suppose instead the covariance matrix were diagonal, $\Sigma = \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}$.

- (a) Are the two variables uncorrelated?
 - (b) Since they are jointly Gaussian, what stronger property follows?
 - (c) What shape are the equal-probability contours?
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Exercise 7 *Maximum likelihood for a spiking neuron.*

A neuron responds on 18 of 30 trials, modelled as Bernoulli(p).

- (a) Write the log-likelihood $\ell(p)$ for k successes in n trials.
 - (b) The MLE is $\hat{p} = k/n$; evaluate it here.
 - (c) Why does maximising the *log*-likelihood give the same answer as maximising the likelihood?
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Exercise 8

- (a) For data assumed Gaussian, the MLE of the mean is the sample mean. For the data $\{2, 4, 9\}$, compute it.
 - (b) In one sentence, explain why fitting a regression line by least squares is a maximum-likelihood method.
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Exercise 9 *Comparing two models by how well they predict the data.*

You observe 8 heads in 10 flips. Model A assumes $p = 0.5$; Model B assumes $p = 0.8$.

- (a) The binomial coefficient $\binom{10}{8}$ is common to both. Using the remaining factors, form the likelihood ratio $\frac{0.8^8 0.2^2}{0.5^{10}}$.
- (b) Which model does the data favour, and by roughly what factor?

Exercise 10 *A psychometric function as a logistic (GLM).*

$\mathbb{P}(\text{detect}) = \sigma(\beta_0 + \beta_1 c)$ with $\beta_0 = -1$, $\beta_1 = 2$, and c the stimulus contrast, where $\sigma(\eta) = 1/(1 + e^{-\eta})$.

- (a) Compute $\mathbb{P}(\text{detect})$ at contrast $c = 0.5$.
 - (b) At what contrast is the detection probability exactly 0.5?
 - (c) Interpret β_1 via the odds (give e^{β_1}).
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Exercise 11 In a logistic model of choice, a predictor has coefficient $\beta = 0.7$.

- (a) Compute the odds ratio e^β .
 - (b) Does the predictor increase or decrease the probability of the outcome?
 - (c) What value of β corresponds to “no effect”?
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Exercise 12 *Model comparison with information criteria.*

Two models are fit to $n = 200$ data points. Model 1: $k = 3$ parameters, $\ln \hat{L} = -410$. Model 2: $k = 6$, $\ln \hat{L} = -405$.

- (a) Compute the AIC of each ($\text{AIC} = 2k - 2 \ln \hat{L}$).
 - (b) Compute the BIC of each ($\text{BIC} = k \ln n - 2 \ln \hat{L}$; use $\ln 200 \approx 5.30$).
 - (c) Which model does each criterion prefer, and why do they disagree?
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Exercise 13 A flexible model achieves almost zero error on its training data but performs poorly on new, held-out data.

- (a) Is this underfitting or overfitting? Is it high bias or high variance?
 - (b) Why is a low *training* error, by itself, not evidence of a good model?
 - (c) What does cross-validation measure that training error cannot?
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Exercise 14 *Information carried by a spiking neuron.*

A neuron’s spike/no-spike output has $\mathbb{P}(\text{spike}) = 0.25$.

- (a) Compute the entropy of its output, in bits.
- (b) Is it more or less than 1 bit? Why?

(c) At what value of p is the entropy maximal?

Exercise 15

- (a) In one sentence, what does the mutual information $I(S; R)$ between a stimulus and a neural response measure?
 - (b) If the response is statistically independent of the stimulus, what is $I(S; R)$?
 - (c) Cross-entropy relates to KL divergence by $H(p, q) = H(p) + D_{\text{KL}}(p||q)$. Since $D_{\text{KL}} \geq 0$, what does this say about the cost of using a wrong model q ?
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Exercise 16

A two-state model of attentional state.

A subject is “engaged” or “distracted”. From engaged, the next trial is 80% engaged, 20% distracted; from distracted, it is 50% engaged, 50% distracted. The subject starts *engaged*, i.e. state vector $(1, 0)$.

- (a) Compute the state distribution after one trial.
- (b) Compute it after two trials.
- (c) State the Markov property in one sentence, and note the link to matrix multiplication.